

# Johan Lakshmanan

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## EDUCATION

### University of Massachusetts, Amherst

*B.S. in Computer Science — GPA: 3.7/4.0, Dean's List*

Amherst, MA

Aug. 2023 – Aug. 2026

## EXPERIENCE

### Roblox

*Software Engineering Intern*

May 2026 – Aug. 2026

San Mateo, CA

- Designed and deployed a versioned bare-metal GPU provisioning pipeline with CI/CD and first-boot validation for **2,400+** GPUs across **300+** nodes, supporting a **\$20M** cloud-to-on-prem fleet migration.
- Reduced GPU node boot-to-ready time by **60%** by pre-baking validated CUDA driver artifacts and implementing backward-compatible health checks, enabling zero-downtime incremental rollout across **10** production clusters.
- Owned production rollout and operational support for critical ML infrastructure, serving production traffic within 4 weeks and preventing a class of SEV0 incidents that caused **5+** hours of model degradation.

### Fidelity Investments

*Quantitative Research Intern*

Jun. 2025 – Aug. 2025

Boston, MA

- Piloted a model-driven trade router, scaling trade volume by **1.4x** and projecting **\$200K+** in annualized savings.
- Developed low-latency routing models and backend services in Python and kdb+/q over **25M+** trades, improving routing selection accuracy by **20%** while keeping decision latency below **10 ms**.
- Engineered scalable ML infrastructure on AWS EKS using Jenkins CI/CD with containerized microservices and MLflow, supporting reproducible model versioning and rollback in production.

### Waters Corporation

*Software Engineering Intern*

May 2024 – Aug. 2024

Milford, MA

- Built a Python-based automated renewal quote system with REST APIs, reducing manual sales-operations workload by **100+** hours per month.
- Designed multiple REST APIs to retrieve customer SharePoint resources by environment ID, cutting incident mitigation times from **hours to minutes** and reducing on-call engineering load.
- Developed C# modules to interface with SAP, optimizing SQL queries and reducing transaction latency by **25%**.

## PROJECTS

### UMass Consulting Group Website | MERN, Redis, AWS

Feb. 2024 – May 2026

- Led full-stack development of a consulting platform using the MERN stack (MongoDB, Express, React, Node.js).
- Rebuilt RESTful APIs with pagination, Redis caching, and performance profiling, reducing latency by **40%** and supporting **5x** higher concurrent request volumes.
- Deployed the MERN platform on Dockerized AWS EC2 with Nginx reverse proxying and GitHub Actions CI/CD.

### Secure Bootloader (MIT Lincoln Lab BWSI) | C, Assembly, Python, AES-GCM

May 2021 – Aug. 2022

- Implemented a secure boot and firmware-update protocol for Stellaris microcontrollers in C/Assembly using AES-GCM authentication and UART communication.
- Built Python penetration-testing tooling to identify and patch **16+** vulnerabilities, earning **1st place** in the MIT Lincoln Laboratory Cybersecurity Challenge.

## SKILLS

**Languages:** Python, C/C++, Java, TypeScript, JavaScript, SQL, C#, kdb+/q

**Cloud & Infra:** AWS (EC2, EKS), Kubernetes, Docker, Terraform, Jenkins, GitHub Actions, Packer, Nginx

**Frameworks & Data:** FastAPI, React, Node.js, Express, Django, PostgreSQL, MongoDB, Redis

**Systems & Tools:** Linux, Git, Bash, CUDA, MIG, VFIO, Distributed Systems, Networking

**AI Developer Tools:** Cursor, Claude Code, Codex CLI

## AWARDS

Hack UMass — **Best ML Hack**

Nov. 2025

Collegiate Penetration Testing Competition — **3rd Place (Global Finals)**

Jan. 2024